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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	SyncEra
Project / Solution Name	Skillogy
Name (Leader)	HAMMADOUCHE Manar
Email & Phone (Leader)	ERA2509065@xmu.edu.my +601121047468
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://syncera-frontend.vercel.app/?utm_source=chatgpt.com
Code Repository URL (public / access granted to judges)	https://github.com/Frank219wan/SyncEra

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The prototype has a React/TypeScript/Vite browser client for Individual career guidance, B1 HR/L&D department planning and B2 TVET/Career Centre analysis. In Real mode, the client calls first-party /api/v1 endpoints through an allowlisted same-origin Vercel proxy to a Python FastAPI backend. FastAPI validates input with Pydantic and produces labelled, deterministic recommendations, skill gaps, workshop and HR planning outputs. A protected Neon PostgreSQL Preview stores synthetic Organization workspaces and analyses with revision compare-and-swap (CAS). Browser localStorage holds draft state; an independent B1 record and Web Locks coordinate single-tab HR editing. Real-mode errors fail closed rather than displaying Demo results. GitHub hosts the source and CI. An optional Tencent TokenHub MiniMax-M3 adapter is implemented for declared synthetic text extraction.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
React 19.1 / React DOM 19.1; React Router DOM 7.6	UI components, routing and stateful views.	MIT
TypeScript 5.8; Vite 6.3; @vitejs/plugin-react 4.4	Typed client, frontend dev/build tooling.	Apache-2.0; MIT
Tailwind CSS 4.1; Lucide React 0.468; @fontsource fonts 5.x	Styling, icons and self-hosted typefaces.	MIT; ISC; font licenses: verify OFL/others
Python 3.12; FastAPI 0.115.12; Pydantic 2.14.1; Uvicorn 0.34.3	HTTP API, schemas, backend runtime.	PSF; MIT; BSD-3-Clause
psycopg 3.2.12 / psycopg-pool 3.3.3; httpx 0.28.1; pytest 8.4.2	PostgreSQL CAS, HTTP integration and tests.	LGPL-3.0; BSD-3-Clause; MIT
PyMuPDF 1.26.7; python-docx 1.2; python-multipart 0.0.32	Synthetic PDF/DOCX intake, multipart processing.	AGPL-3.0 or commercial ; MIT; Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Tencent TokenHub / MiniMax-M3 (optional)	Optional server-side chat-completions for synthetic text extraction; not used in reported deterministic B1/B2 tests.	free
SyncEra first-party /api/v1 (not external)	Frontend proxy to FastAPI for profile, radar, B1/B2 analyses and results.	Self-hosted internal API
DOSM data publications (no active live API)	Public evidence ingested as files into candidate local snapshots; not a runtime third-party API.	Public source; publisher terms
TalentCorp MyCOL (not a live API)	Public research/evidence material for a staged candidate snapshot.	Public publication; publisher terms
Other runtime third-party APIs	B1/B2 deterministic path.	Free

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel — existing frontend project	Hosts protected React Preview and allowlisted same-origin API proxy; Vercel Authentication enabled.	User-side;frontend
Vercel — existing backend project	Hosts FastAPI serverless Preview; runtime secrets remain server-side.	Client side;backend
Neon — PostgreSQL	Isolated Preview relational database for synthetic Organization workspaces / CAS analyses.	Client side;backend
GitHub + GitHub Actions	Code repository, PR review, automated frontend/backend/integration CI.	Repository + hosted CI; whole code
Browser localStorage + Web Locks (not cloud)	Device-local draft persistence and B1 cross-tab coordination.	Client-side only; backend

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
MiniMax-M3 (optional integration)	Tencent TokenHub	Pre-trained model; opt-in for synthetic text only	Tencent/provider terms; check from tencent
B1 HR analysis (organization-b1-v0.1)	SyncEra team	Deterministic rules combined with AI model	Team-owned source; repo Apache-2.0
B2 Career Centre (organization-b2-v0.1)	SyncEra team	Deterministic rules combined with AI model	Team-owned source; repo Apache-2.0
Individual matching (tech-v1 / rules-v0.1)	SyncEra team	Deterministic rules combined with AI model	Team-owned source; repo Apache-2.0
Fine-tuned / self-trained ML models	None	None	None

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic role/skill catalogue + fixtures	Repo: backend/app/catalogue.py; src/data.ts	Self-created synthetic (active)	Team-created; repo Apache-2.0	N
Synthetic B1/B2/Individual QA records	Repo: tests/; synthetic Preview QA	Synthetic evaluation / fixtures	Team-created; repo Apache-2.0	N
DOSM JMI + Employment Statistics, Q2 2026	dosm.gov.my; backend/data/dosm_market_q2_2026.json	Public (staged / feature-gated)	DOSM reuse/attribution; verify terms	N
TalentCorp MyCOL 2024/25 research	talentcorp.com.my/mycol/; candidate snapshot	Public report (candidate only)	Publisher reuse terms; verify	N
Microsoft Learn course/resource candidates	learn.microsoft.com; backend/tech_delivery/	Public links / resource metadata	Microsoft content terms; verify	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI ChatGPT (model/version varies)	Architecture/PRD drafts, technical discussion, QA and declaration preparation.	Major
OpenAI Codex (agent/CLI; version varies)	Code generation/refactoring, unit tests, regression debugging, CI/release procedures.	Major
Other GenAI development tools	Doubao	Minor
Visual design/image GenAI	Doubao	Minor
Other	None	None

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

ChatGPT and Codex outputs were reviewed through human-led code inspection, independent QA, regression tests, TypeScript checks, builds, cross-stack tests and GitHub Actions CI. The team tested deterministic outputs with fictional data and validated error, revision/CAS and multi-tab scenarios. AI-generated suggestions were not treated as verified facts, production-ready changes, or a substitute for formal independent review. Unresolved Preview failures and limits are recorded explicitly; the team owns all submitted code and declarations.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Credit in Replit platform	Perform a global deployment test
Others	None	None
—	—	—
—	—	—

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No, we use it to improve some functionalities in the frontend.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The intended hackathon workflow uses fictional/synthetic profiles, HR departments, optional anonymous employee records and fictional TVET student records; real personal submissions are not authorized. Browser drafts persist in device-local localStorage until the user clears site data. Real-mode Organization requests send synthetic input to a protected FastAPI service, whose isolated Neon Preview database stores workspace data and analyses for testing. Vercel Preview access uses Vercel Authentication; API traffic is proxied through an allowlist and provider credentials must stay server-side. Test-created database records were manually verified and precisely cleaned up in reported runs; Real CVs, student or employee details must not be entered.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Recommendations and HR training priorities are rule-based guidance, not validated forecasts, hiring decisions or employee assessments. Market evidence may be synthetic, dated, coarse, or pending independent source approval. Unknown employee skill baselines are marked unknown rather than zero. Optional MiniMax extraction may hallucinate; output is schema-validated and requires human confirmation. The B1/B2 system uses optimistic CAS but cannot safely recover the original server revision after a lost response without an authenticated readback design; fallback creation requires explicit user consent. Multi-tab writes are Web Locks guarded and fail closed if locking is unavailable. Latest lifecycle/CAS fixes still require protected-Preview verification; independent review, actual backend-fault testing and historical log audit remain incomplete. PyMuPDF AGPL/commercial-license compliance must be checked before broader deployment.

7. Compliance & Readiness Checklist

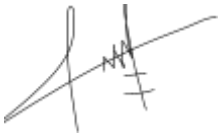
Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]
24/10/2026